

# OpenWebSearch.eu — An Open Scalable Infrastructure for Web Search & AI

Prof. Dr. Michael Granitzer Chair of Data Science, University of Passau Coordinator  
OpenWebSearch.eu



# Web Search Today

A critical resource managed as an oligopoly

## Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

**1.15 Billion**

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

**Top 0.1%–1% of sites**

may account for roughly half of all pages  
heavy-tail intuition from large web graph studies

**~ 400 Billion**

documents in Google's search index  
2020 figure cited in US v. Google testimony

*Sites ≠ domains ≠ pages; use exact counts cautiously*

### VELOCITY

**+8.6 M / month**

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

**~ 3.2 / second**

same monthly net change, averaged across December

**149 ZB (global)**

all digital data created / copied worldwide in 2024  
IDC / Statista global datasphere estimate

*The web changes constantly; crawling is continuous, not one-off*

### VARIETY

**150+ languages**

observed across websites

W3Techs usage survey

**249 CMS Platforms**

CMS platforms identified in the HTTP Archive dataset  
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

**HTML · PDF · JSON-LD · media · APIs**

content types to parse and process

*HTML · PDFs · JS apps · schemas · media · APIs*

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) · US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista



- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

**1.15 Billion**

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

**Top 0.1%–1% of sites**

may account for roughly half of all pages  
heavy-tail intuition from large web graph studies

**~ 400 Billion**

documents in Google's search index  
2020 figure cited in US v. Google testimony

*Sites ≠ domains ≠ pages; use exact counts cautiously*

### VELOCITY

**+8.6 M / month**

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

**~ 3.2 / second**

same monthly net change, averaged across December

**149 ZB (global)**

all digital data created / copied worldwide in 2024  
IDC / Statista global datasphere estimate

*The web changes constantly; crawling is continuous, not one-off*

### VARIETY

**150+ languages**

observed across websites

W3Techs usage survey

**249 CMS Platforms**

CMS platforms identified in the HTTP Archive dataset  
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

**HTML · PDF · JSON-LD · media · APIs**

content types to parse and process

*HTML · PDFs · JS apps · schemas · media · APIs*

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·  
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista



- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

High upfront costs to tap the Web as resources  
— especially small innovators and researchers  
are left behind

# But does it matter? Yes — on four dimensions

## User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

## Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

## Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

## AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

Being able to navigate the Web is as fundamental as GPS or the power grid

# OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

## Four Objectives

- 1. Open Technology Stack** Open-source pipelines for crawling, preprocessing, indexing
- 2. Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
- 3. Added Value Services** Dashboard, tools, data access APIs
- 4. Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

## 14 Core Partners



Funded by **Horizon Europe** (NGI programme)

# The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

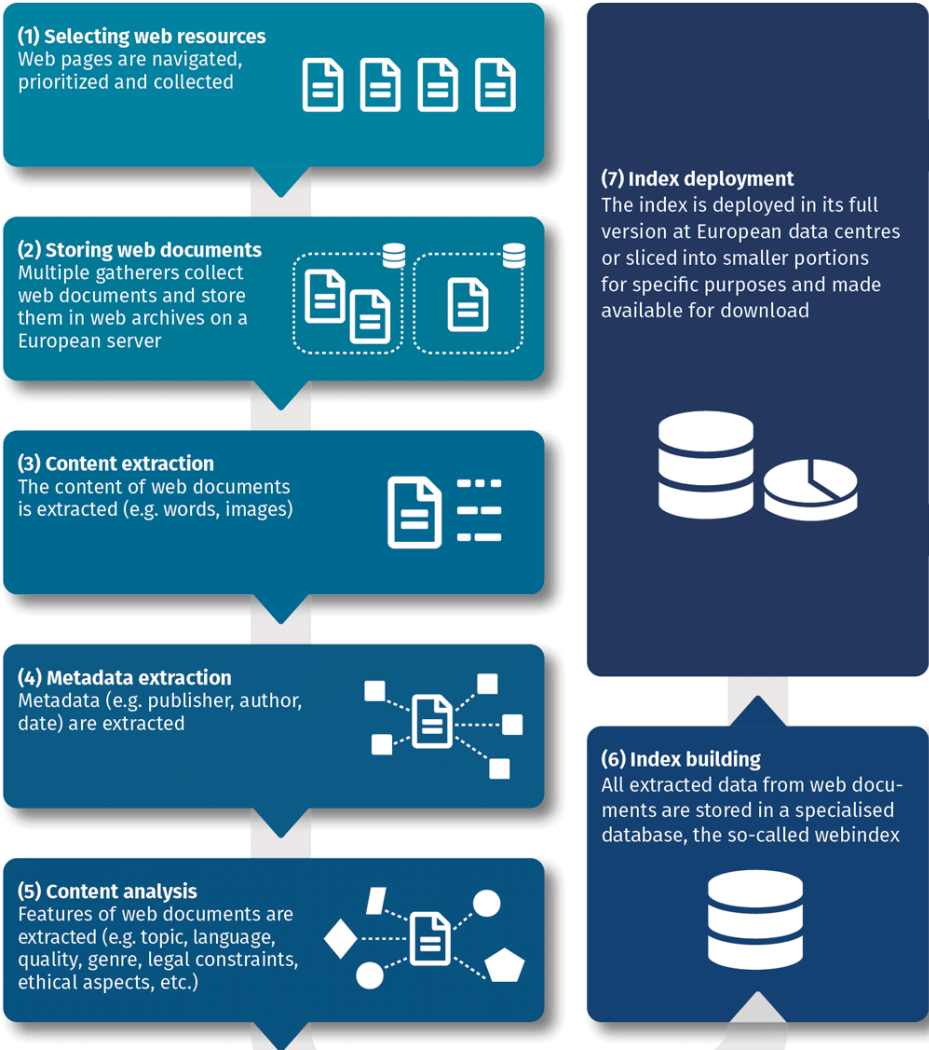
**Our Proposition:** A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe



# Conceptual Workflow

## Index Generation

Web resources are selected and retrieved, their content and metadata are analysed, and all data stored in the index database.



## Search Applications

A user search request will be answered by a search application that makes use of the open web index.



## Data Products

Knowledge representation models will be created using the open web index, in order to be used by any agent and for many applications





# Daily Workflow Pipeline



## Crawlig Infrastructure (2026)

- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

## Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement

# Accessing the Open Web Index

# Dashboard — Statistics & Datasets in Detail



## OWI Statistics Tab

- Repository health per data center (IT4I · LRZ)
- Key indicators: pipeline runs, HPC core-hours consumed, avg. duration, active pipelines
- Crawled data (1.3 PB), WARC datasets, public datasets, index size
- Language distribution chart + crawl volume timeline per DC
- Dataset collection distribution by resource type (main / GPU / curlie / embeddings)

## OWI Datasets Tab

- Filter: datacenter · format (WARC / Parquet / CIFF / Embeddings) · language · date range
- Infinite-scroll dataset cards: title, date, DC, public/private badge, file listing
- Parquet file preview in-browser; download via LEXIS Portal or owilix remote pull

# Dashboard — Corpora, Models & Search

## OWI Corpora

Curated, administrator-prepared collections for distribution:

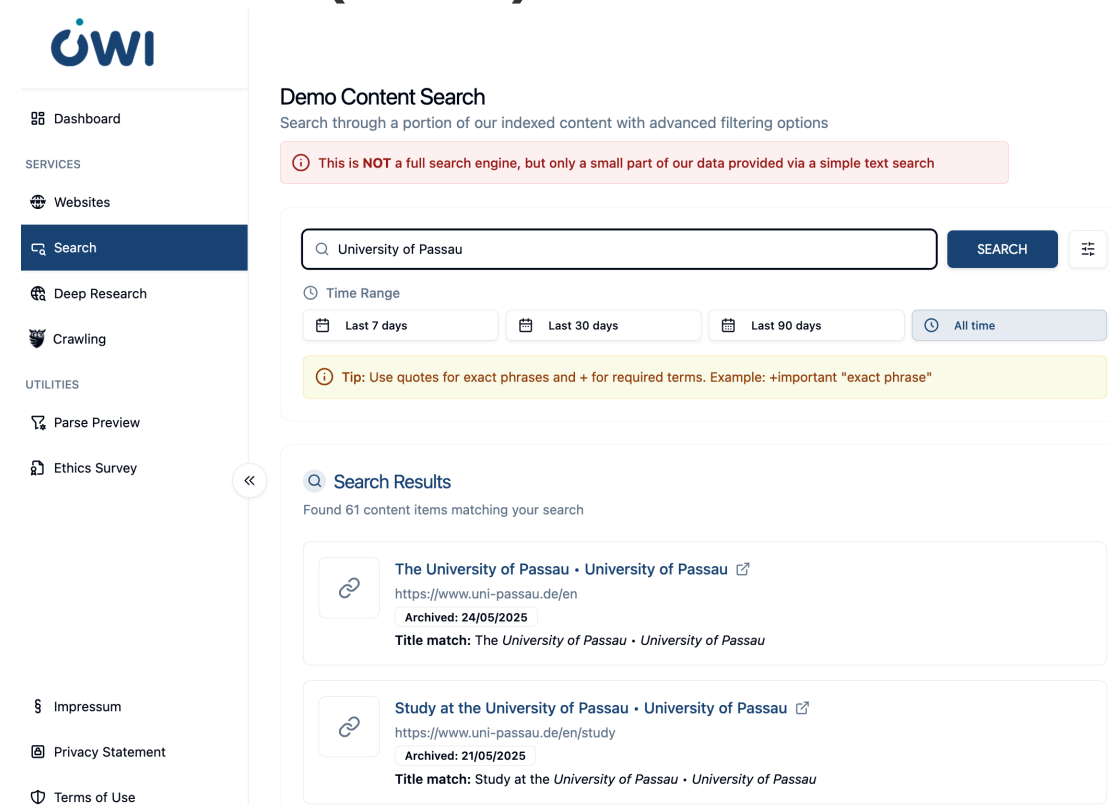
- Licensing metadata per corpus (creator, terms, full licence text)
- LEXIS integration for storage, access control and metadata
- Includes domain-specific and project-specific collections

## OWI Models

Registry of language models for web search:

- Browser-runnable models via WebAssembly (Wllama library)
- Cloud-based models (OpenAI, DeepSeek) for heavy tasks
- Model cards with training data, benchmarks,  and GGUF file listing

## OUR Search (Demo)



The screenshot shows the OWI Search Demo interface. On the left is a sidebar with the OWI logo and a navigation menu including Dashboard, Websites, Search (highlighted), Deep Research, Crawling, Parse Preview, and Ethics Survey. The main content area is titled 'Demo Content Search' and includes a search bar with 'University of Passau' entered, a 'SEARCH' button, and a 'Time Range' filter with options for 'Last 7 days', 'Last 30 days', 'Last 90 days', and 'All time'. A tip box suggests using quotes for exact phrases and '+' for required terms. Below the search bar, the 'Search Results' section shows 'Found 61 content items matching your search'. Two result cards are visible, each with a thumbnail, title, URL, archive date, and title match.

**OWI**

Dashboard

SERVICES

Websites

**Search**

Deep Research

Crawling

UTILITIES

Parse Preview

Ethics Survey

Impressum

Privacy Statement

Terms of Use

### Demo Content Search

Search through a portion of our indexed content with advanced filtering options

**ⓘ** This is **NOT** a full search engine, but only a small part of our data provided via a simple text search

University of Passau **SEARCH**

Time Range

Last 7 days Last 30 days Last 90 days All time

**ⓘ** Tip: Use quotes for exact phrases and + for required terms. Example: +important "exact phrase"

### Search Results

Found 61 content items matching your search

 **The University of Passau • University of Passau**   
https://www.uni-passau.de/en  
Archived: 24/05/2025  
Title match: The University of Passau • University of Passau

 **Study at the University of Passau • University of Passau**   
https://www.uni-passau.de/en/study  
Archived: 21/05/2025  
Title match: Study at the University of Passau • University of Passau

Simple + advanced field-specific search over a subset of the index; time-range filtering; result cards with thumbnail, title, URL

## Deep Research (Experimental)

AI-powered multi-step research: configurable depth + breadth, local or cloud LLM, interactive clarification mode

# owilix — The Open Web Index Client



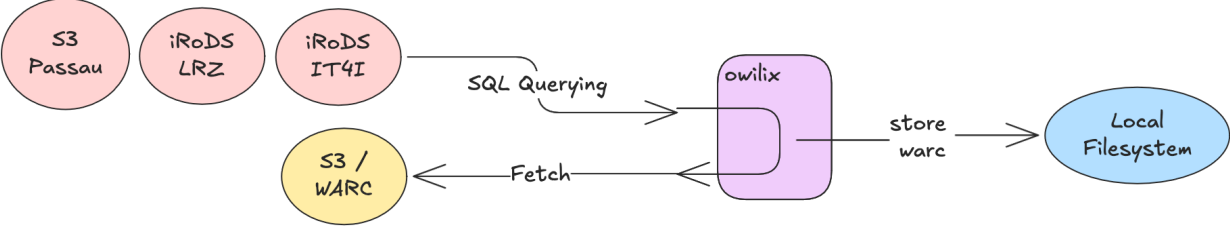
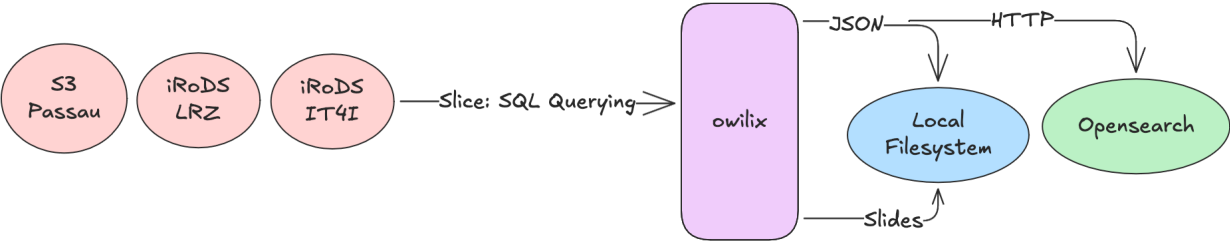
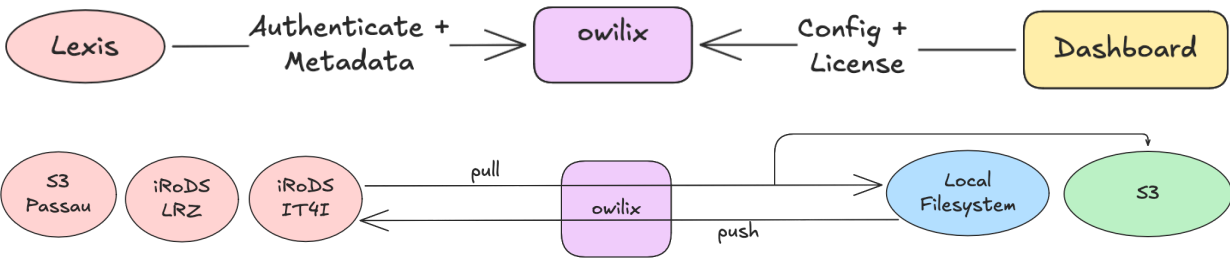
Four main workflows:

1. **Authentication** — B2ACCESS via LEXIS + Dashboard config (OpenID Connect / Keycloak)
2. **Sync Datasets** — pull/push between federated iRODS/S3 storage and local filesystem
3. **Query / Slice** — client-side SQL queries with JSON, HTTP, OpenSearch output
4. **WARC Access** — fetch original WARC files via SQL querying

Slice the index by:

- **Domain** (.de, .at, .fr, ...)
- **Topic** (Curlie.org hierarchy labels)
- **Language** (filename-based selection)
- **Sitelist** (CSV of URLs / domains / TLDs)
- **Structured data** / microdata type
- **Outlinks** for link graph analysis

# owilix Workflows



# Remote Index — Query Without Downloading

## What is the remote index?

A DuckDB / Apache Spark-queryable representation of the OWI — Parquet partitions aligned with CIFF shards, accessible from remote clients without downloading the full index.

## Example queries

```
-- Count documents per language
SELECT language, COUNT(*) as docs
FROM read_parquet('remote://lrz:latest/main/**')
GROUP BY language ORDER BY docs DESC;

-- Filter by domain
SELECT url, title FROM ... WHERE domain = 'wikipedia.org';
```

## Use cases

- Analytics and statistics on the live index
- Lightweight RAG pipeline ingestion
- Ad-hoc exploration before committing to a full  shard download

## OUR<sup>2</sup>S — Online Search API

OUR<sup>2</sup>S (built on Vespa) extends the remote index with a full REST search API:

- Structured document model: full page · snippets · embeddings
- Keyword, semantic and hybrid ranking modes
- Multi-index federation across data centers
- No local index needed — query directly over network

→ OUR<sup>2</sup>S details in the dedicated OUR<sup>2</sup>S column

## Access summary

| Mode   | Tool      | Authentication       |
|--------|-----------|----------------------|
| Browse | Dashboard | Anonymous / B2ACCESS |

| Mode                  | Tool                   | Authentication      |
|-----------------------|------------------------|---------------------|
| Download              | owilix                 | B2ACCESS<br>(LEXIS) |
| SQL queries           | DuckDB<br>remote       | B2ACCESS            |
| REST search           | OUR <sup>2</sup> S API | API key             |
| Conversational<br>RAG | MosaicRAG              | Open /<br>B2ACCESS  |

# MosaicRAG — Building RAG Systems on OWI

## From OWI Datasets to Conversational Search

MosaicRAG consumes OWI daily datasets directly — CIFF inverted index files, Parquet metadata records, and dense embeddings (published since Dec 2025) — and exposes them through a modular conversational interface.

### Pipeline for building your own MosaicRAG instance:

1. Download OWI slice with `owilix` (topic, domain, or language filter)
2. Ingest CIFF → inverted index; Parquet → DuckDB; embeddings → ChromaDB
3. Configure retrieval pipeline: sparse, dense, or hybrid
4. Choose reranker (TF-IDF · Embedding cross-encoder · Tournament LLM · Group LLM)
5. Optionally add summarization and relevance marking

Full documentation: [openwebindex.eu/book](https://openwebindex.eu/book) — "Building vertical search engines with MOSAIC"

### Supported LLMs

| Task              | Models                    |
|-------------------|---------------------------|
| Reranking         | gemma2, qwen2.5, llama3.1 |
| Summarization     | + DeepSeekv3              |
| Relevance marking | gemma2, qwen2.5, llama3.1 |

### Live Demo

Try conversational search on the Open Web Index at [mosaicrag.ows.eu](https://mosaicrag.ows.eu)

# Using the Open Web Index — Applications

## Realised Applications

- **Mosaic & MosaicRAG** — modular vertical search framework; science, health, arts demos → [mosaic.ows.eu](https://mosaic.ows.eu)
- **Earth Observation Search** — RAG QA + geo-contextualised search (DLR / Radboud)
- **Multilingual Scientific Index** — enriching Finland's Research.fi portal (CSC)
- **Restaurant Recommender** — privacy-preserving mobile app; 214-user pilot (A1 Slovenia)
- **Argumentation Search** — args.me extended with OWI pro/contra index →

## Community Applications (FSTPs)

- **AKSE** — argumentation knowledge graphs
- **VERITAS / DEXAI** — verification-oriented RAG
- **OMMS** — enriching OpenStreetMap with POI metadata
- **DTCommerce** — SME e-commerce product enrichment
- **CIFFIL** — cross-collection index for municipal search
- **TILDE** — trustworthy health domain search

The OWI enables a full ecosystem of vertical search engines, RAG systems,

[args.me/owi](https://args.me/owi)

- **CERN Search** — institutional search + disability knowledge search (Nooon)

and domain-specific applications

# Using the Open Web Index — Data & Analytics

## Published Datasets

| Dataset         | Scale                 | Use case                        |
|-----------------|-----------------------|---------------------------------|
| WebFAQ          | 96M Q&A, 75 langs     | QA training, cross-lingual IR   |
| WebFAQ 2.0      | 198M pairs, 108 langs | Dense retrieval, hard negatives |
| Imprints        | 5.25M hosts, geocoded | Enterprise analytics, Eurostat  |
| German Commons  | 154B tokens           | German LLM pre-training         |
| OWS-Curlie-2025 | 20.7M docs + qrels    | IR evaluation benchmark         |

All datasets available at [openwebindex.eu/corpora](https://openwebindex.eu/corpora)

## Analytics Potential

- **Official Statistics:** enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring:** supply chain analysis, news monitoring (JRC)
- **AI Training Data:** domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks:** web-scale IR test collections with relevance assessments
- **Trend Analysis:** temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

# In Summary

## What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

## Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

# Before we come to the Questions

# We are looking for ...

## Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

## Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

# Get in touch

|                        |                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------|
|                        |                                                                                                                 |
| <b>Website</b>         | <a href="https://openwebsearch.eu">openwebsearch.eu</a>                                                         |
| <b>Dashboard</b>       | <a href="https://openwebindex.eu">openwebindex.eu</a>                                                           |
| <b>Code Repository</b> | <a href="https://opencode.it4i.eu/openwebsearcheu-public/">https://opencode.it4i.eu/openwebsearcheu-public/</a> |
| <b>Community</b>       | <a href="https://openwebsearch.eu/community">openwebsearch.eu/community</a>                                     |
| <b>Join</b>            | <a href="mailto:join@openwebsearch.org">join@openwebsearch.org</a>                                              |
| <b>General</b>         | <a href="mailto:community@openwebsearch.eu">community@openwebsearch.eu</a>                                      |

Thank you — Questions, Comments, Feedback very welcome!